

Roles of Diffusion

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graph TD; A[Roles of Diffusion] --- B[Oxidation]; A --- C[Sintering]; A --- D[Doping]; A --- E[Many more...]; A --- F[Creep]; A --- G[Aging]; A --- H[Carburizing]; A --- I[Material Joining]; A --- J[Phase Transformation]; B --- B1[Metals]; F --- F1[Many mechanisms]; I --- I1[Diffusion bonding];
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Oxidation

Metals

Sintering

Doping

Semiconductors

Many more...

Creep

Many mechanisms

Aging

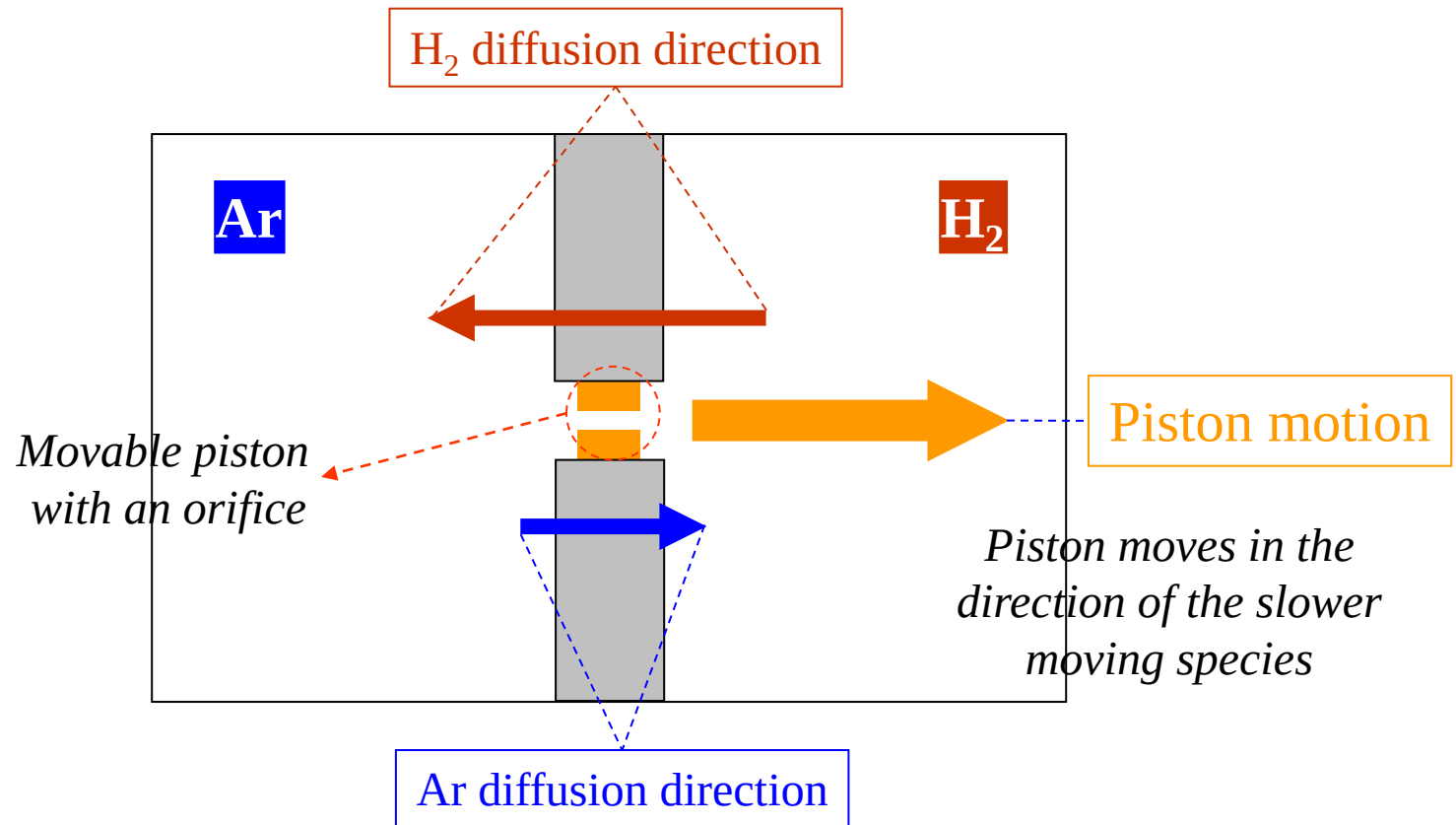
Precipitates

Carburizing

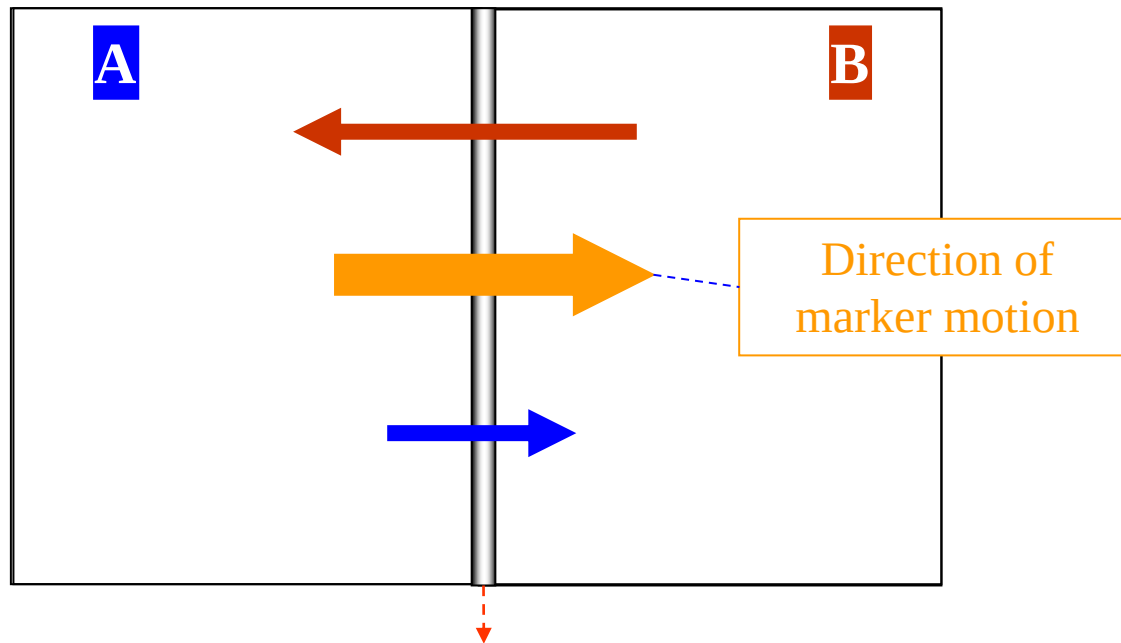
Steels

Material Joining *Diffusion bonding*

Phase Transformation



Kirkendall effect



Inert Marker is basically a thin rod of a high melting material, which is insoluble in A & B

Diffusion

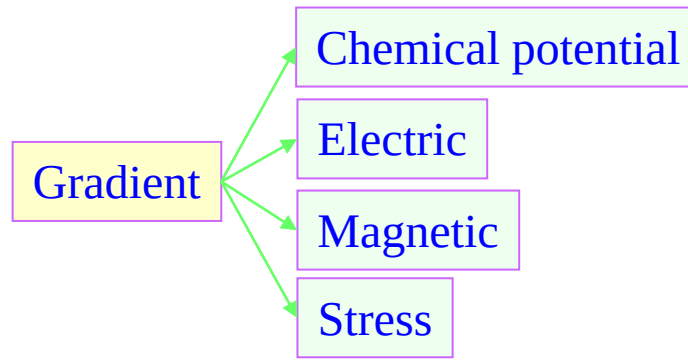
Gradient

Chemical potential

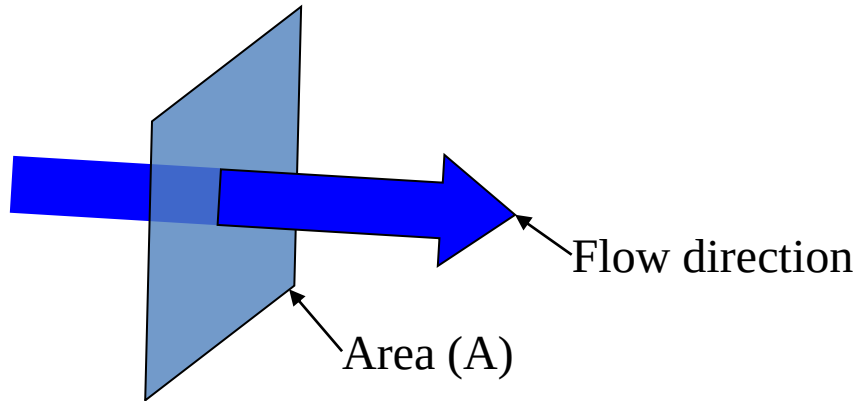
Electric

Magnetic

Stress

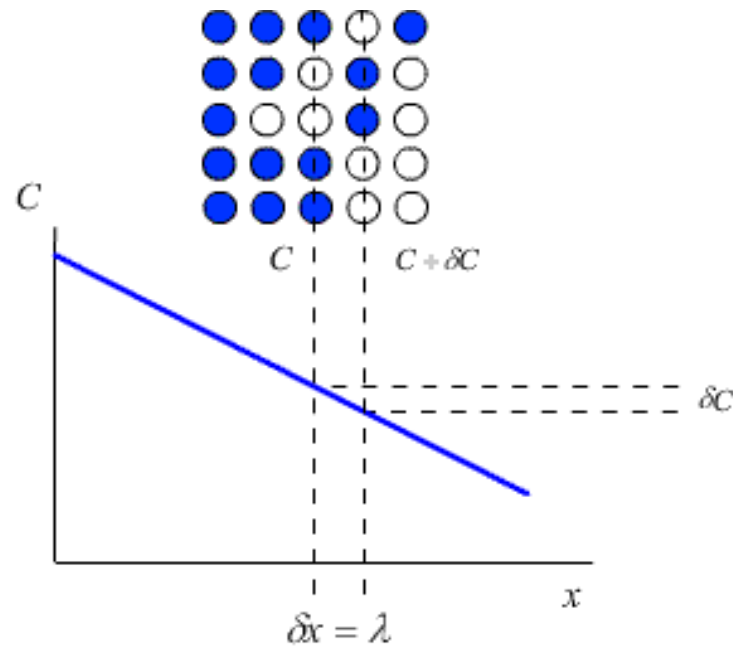


Important terms



$$J = \frac{1}{A} \frac{dn}{dt}$$

$$J \rightarrow \left[\frac{\text{mass}}{\text{area} \cdot \text{time}} \right] = \left[\frac{\text{atoms}}{\text{m}^2 \cdot \text{s}} \right]$$



Fick's I law

The negative sign implies that diffusion occurs down the concentration gradient

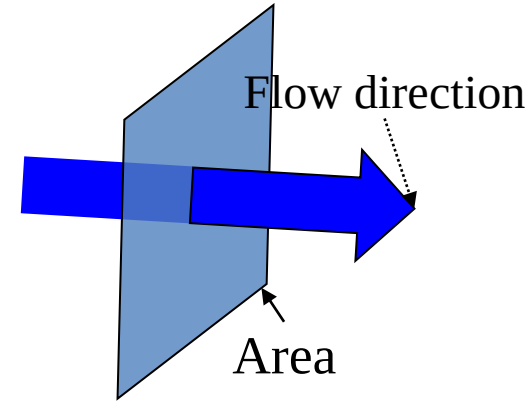
A material property

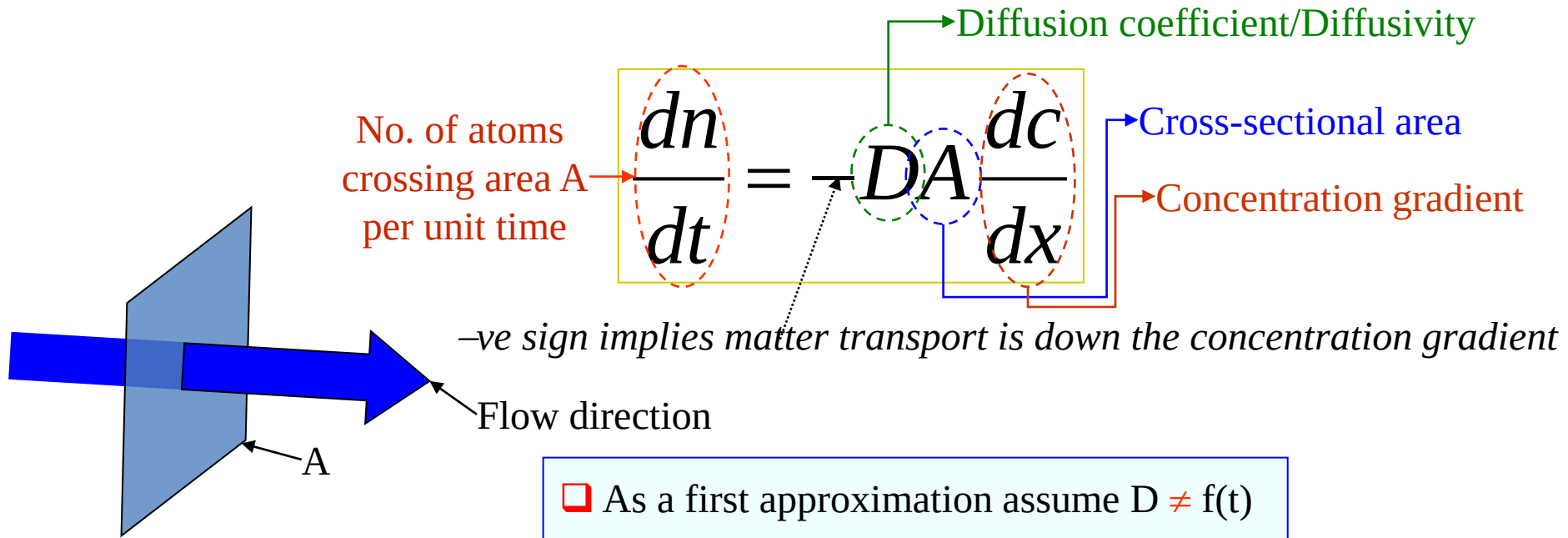
$$J \propto \frac{dc}{dx}$$

$$J = -D \frac{dc}{dx}$$

$$J = \frac{1}{A} \frac{dn}{dt} = -D \frac{dc}{dx}$$

$$\frac{dn}{dt} = -DA \frac{dc}{dx}$$





Let us look at the units of Diffusivity

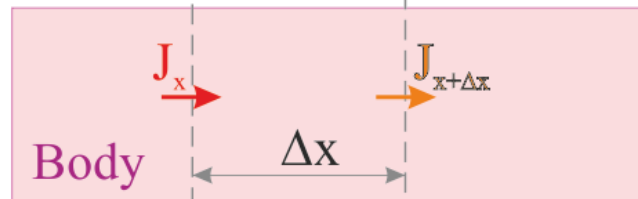
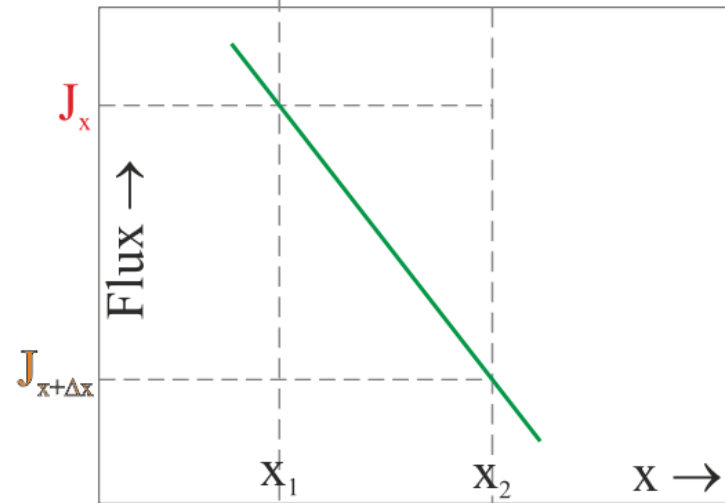
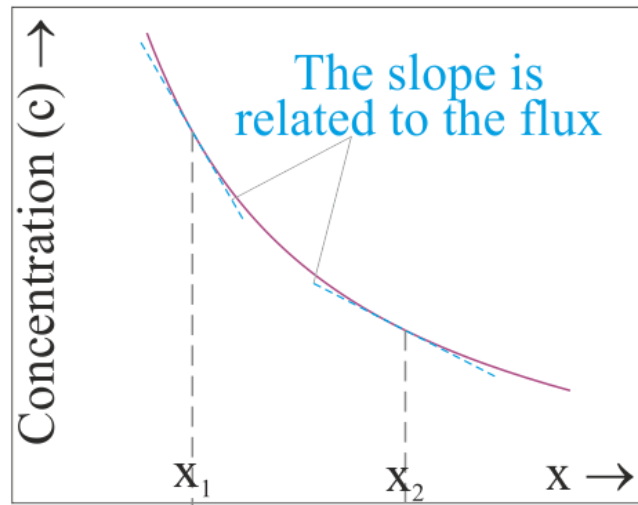
$$\left\{ J = -D \frac{dc}{dx} \right\} \rightarrow \left[\frac{\text{number}}{m^2 s} \right] = [D] \left[\frac{\text{number}}{m^3} \cdot \frac{1}{m} \right]$$

$$\left[\frac{m^2}{s} \right] = [D]$$

Fick's II law

$$\left(\frac{\partial c}{\partial t}\right) = \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right)$$

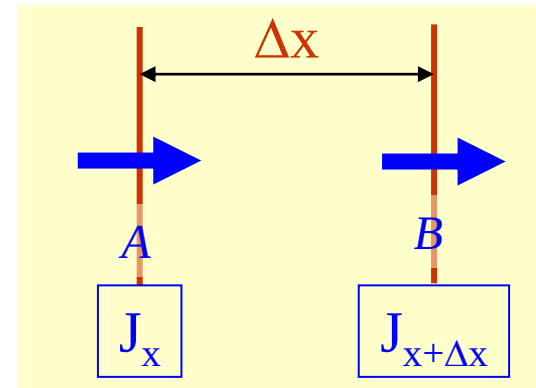
$$\left(\frac{\partial c}{\partial t}\right) = D \frac{\partial^2 c}{\partial x^2}$$



$$Accumulation = J_x - J_{x+\Delta x}$$

$$Accumulation = J_x - \left\{ J_x + \frac{\partial J}{\partial x} \Delta x \right\}$$

$$\left(\frac{\partial c}{\partial t} \right) \Delta x = J_x - \left\{ J_x + \frac{\partial J}{\partial x} \Delta x \right\}$$



$$\left[\left(\frac{Atoms}{m^3} \frac{1}{s} \right) \cdot m \right] = \left[\frac{Atoms}{m^2 s} \right] \equiv [J]$$

Calculation of units

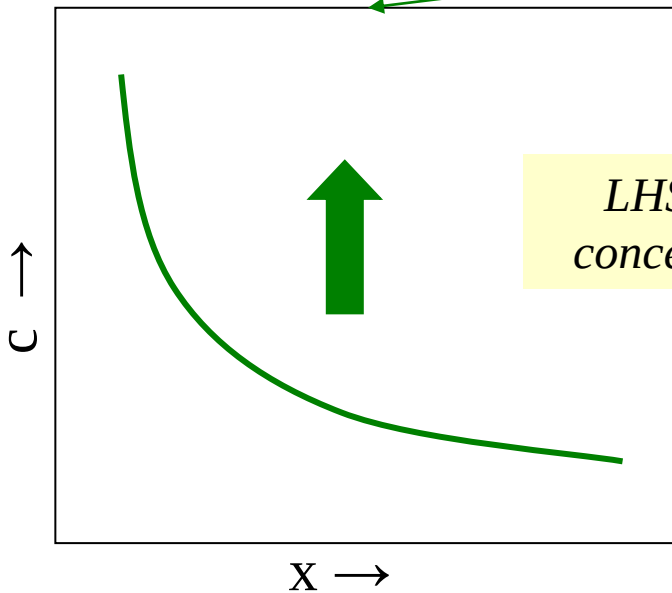
$$\left(\frac{\partial c}{\partial t} \right) \Delta x = - \frac{\partial J}{\partial x} \Delta x \rightarrow \left(\frac{\partial c}{\partial t} \right) = - \frac{\partial}{\partial x} \left(- D \frac{\partial c}{\partial x} \right) \rightarrow \text{Fick's first law}$$

$$\left(\frac{\partial c}{\partial t} \right) = \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right) \xrightarrow{D \neq f(x)} \left(\frac{\partial c}{\partial t} \right) = D \frac{\partial^2 c}{\partial x^2}$$

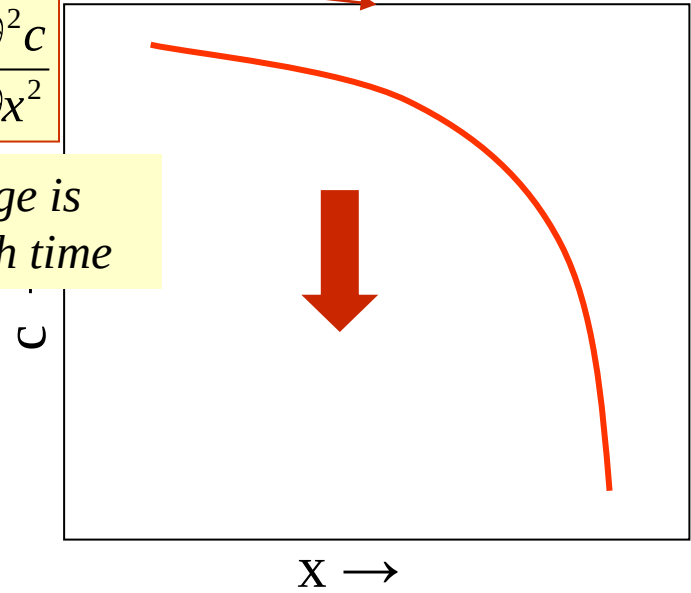
RHS is the curvature of the c vs x curve

$$\left(\frac{\partial c}{\partial t}\right) = D \frac{\partial^2 c}{\partial x^2}$$

LHS is the change in concentration with time

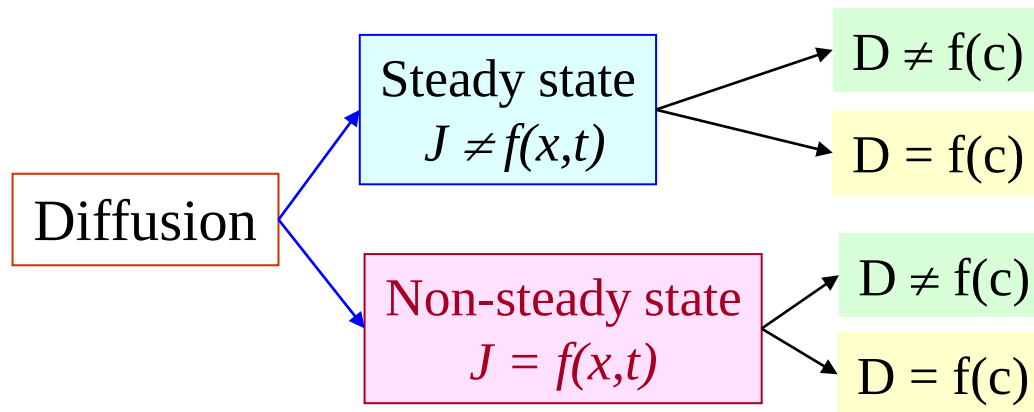


+ve curvature $\Rightarrow c \uparrow$ as $t \uparrow$



-ve curvature $\Rightarrow c \downarrow$ as $t \uparrow$

Steady and non-steady state diffusion



$$\left(\frac{\partial c}{\partial t}\right) = \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right)$$

Under steady state conditions

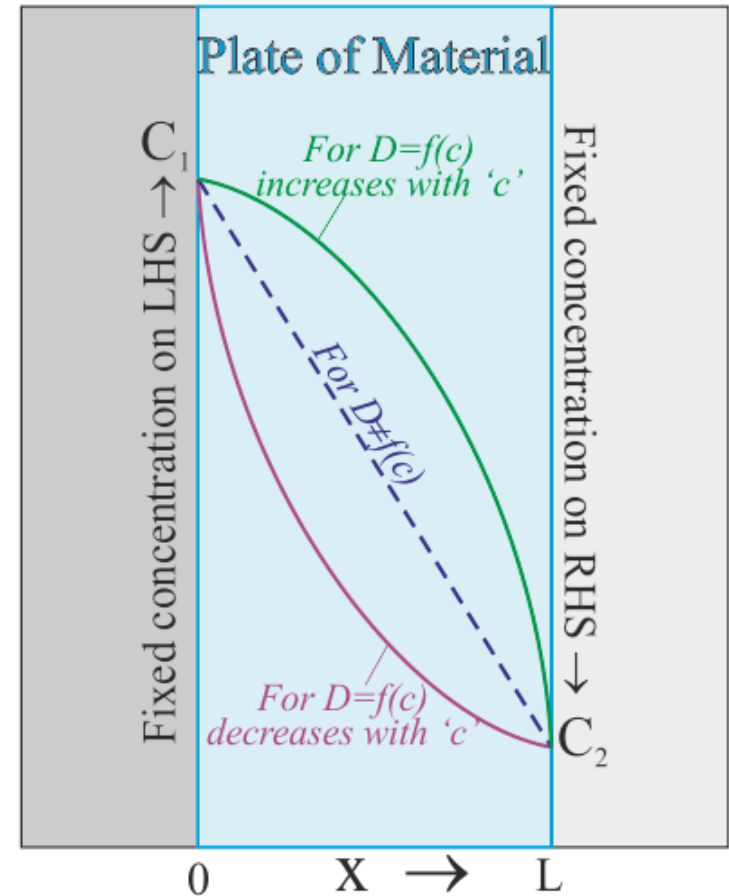
$$\left(\frac{dc}{dt}\right)_x = -\left(\frac{\partial J}{\partial x}\right)_t = 0$$

Substituting for flux from Fick's first law

$$-\frac{\partial}{\partial x} \left(-D \frac{\partial c}{\partial x} \right) = 0 \quad \xrightarrow{\text{If } D \text{ is constant}} \quad D \frac{\partial^2 c}{\partial x^2} = 0$$

If D is NOT constant

$$D \frac{\partial c}{\partial x} = \text{constant}$$

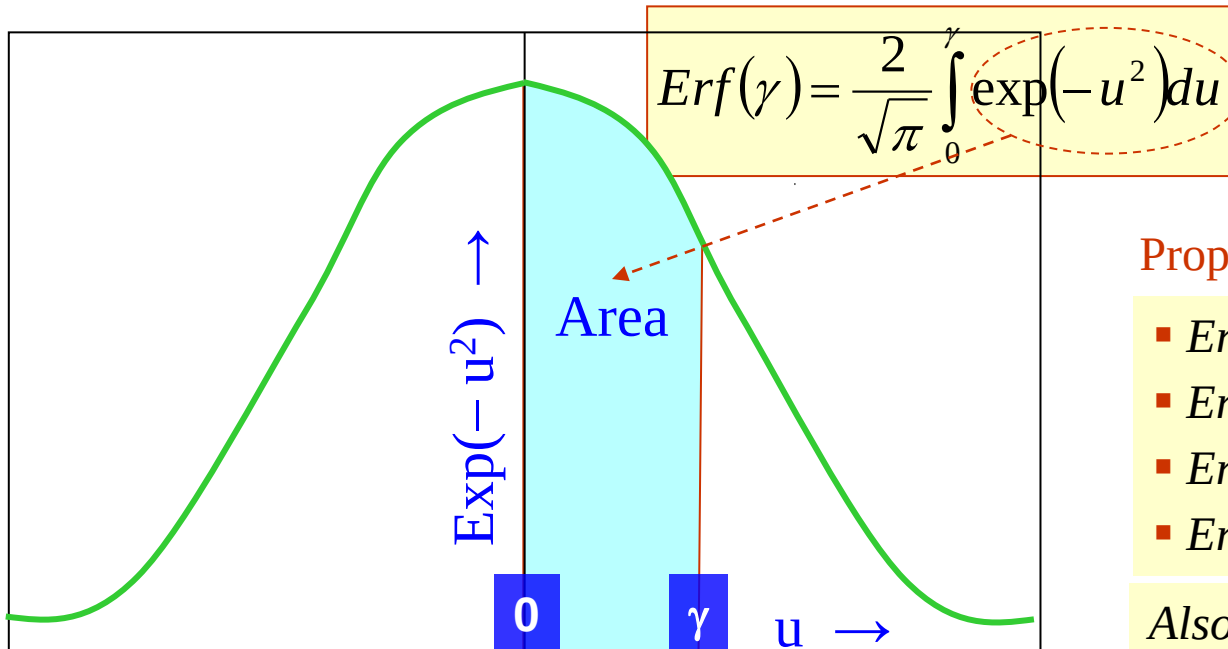


Under non-steady state conditions

$$\left(\frac{\partial c}{\partial t}\right) = \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right) \xrightarrow{\text{If } D \text{ is not a function of position}} \left(\frac{\partial c}{\partial t}\right) = D \frac{\partial^2 c}{\partial x^2}$$

- ❑ The first simplification we make for the non-steady state conditions is that ‘D’ is not a function of the position.
- ❑ If the diffusion distance is short relative to dimensions of the initial inhomogeneity, we can use the **error function** (*erf*) solution with 2 arbitrary constants.
- ❑ The constants can be solved for from Boundary Condition(s) and Initial Condition(s).

$$c(x, t) = A - B \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right)$$



Properties of the error function

- $\text{Erf}(\infty) = 1$
- $\text{Erf}(-\infty) = -1$
- $\text{Erf}(0) = 0$
- $\text{Erf}(-x) = -\text{Erf}(x)$

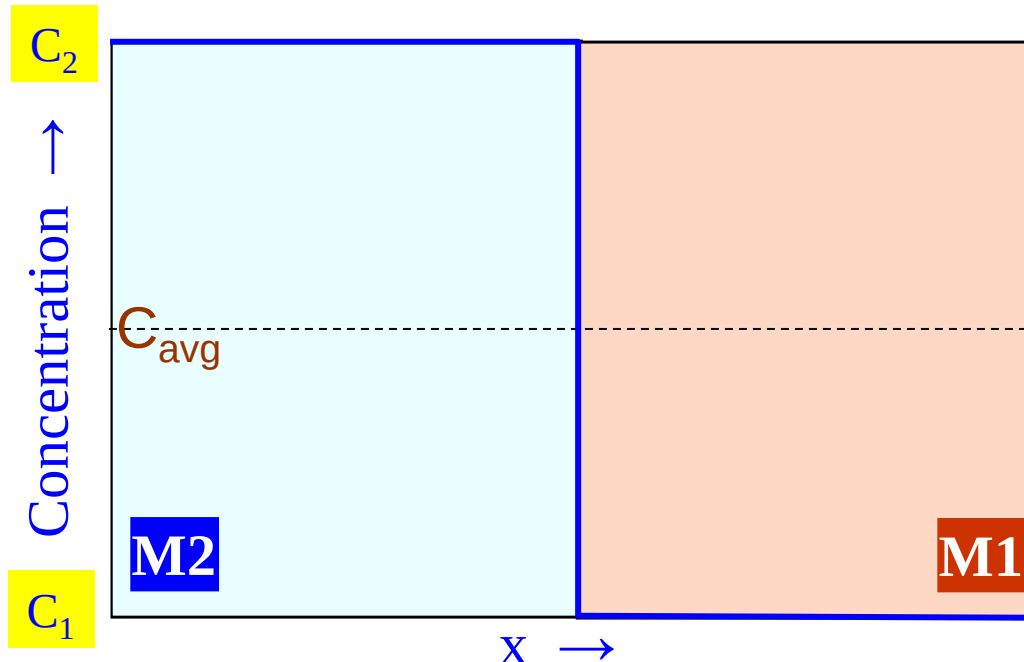
Also

- For upto $x \sim 0.6 \rightarrow \text{Erf}(x) \sim x$
- $x \rightarrow 2, \text{Erf}(x) \rightarrow 1$

- ✓ Let two materials **M1** & **M2** be joined together and kept at a temperature (T_0), where diffusion is appreciable. Let C_1 be the concentration of a species in M1 and C_2 in M2.
- ✓ This is a 1D diffusion problem (i.e. the species diffuses along x-direction only).
- ✓ The initial concentration profile (at $t = 0$, $c(x,0)$) of a species is like a step function (blue line). If M1 and M2 are pure materials, then C_1 would be zero.
- ✓ We can define an average composition of the species as: $(C_1 + C_2)/2$.

The *initial conditions* (at $t = 0$) can be written as:

- $C(+x, 0) = C_1$
- $C(-x, 0) = C_2$




$$t_1 > 0 \mid c(x, t_1)$$

t = 0 | c(x,0)

Concentration $\rightarrow$

$$C_1$$
 C_{avg}

M2

M1

X →

 $f(x)|_t$ $f(t)|_x$

Non-steady state

- If $D = f(c)$

$$\Rightarrow c(+x,t) \neq c(-x,t)$$

i.e. asymmetry about y-axis

$$c(x,t) = A - B \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

- $C(+x, 0) = C_1$
- $C(-x, 0) = C_2$

- $A-B = C_1$
- $A+B = C_2$

- $A = (C_1 + C_2)/2$
- $B = (C_2 - C_1)/2$

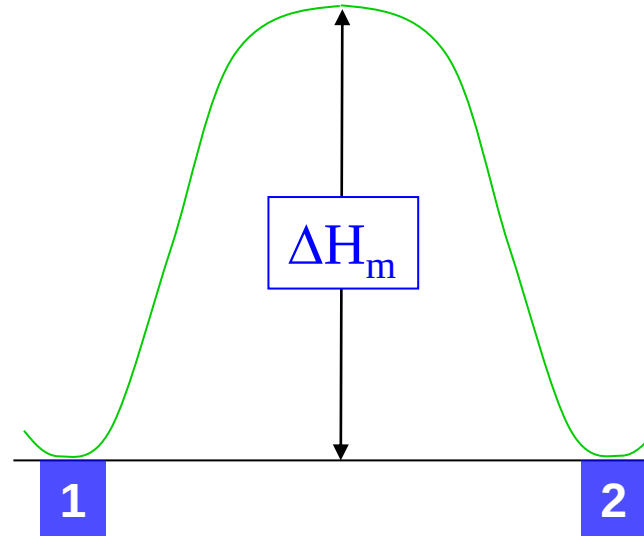
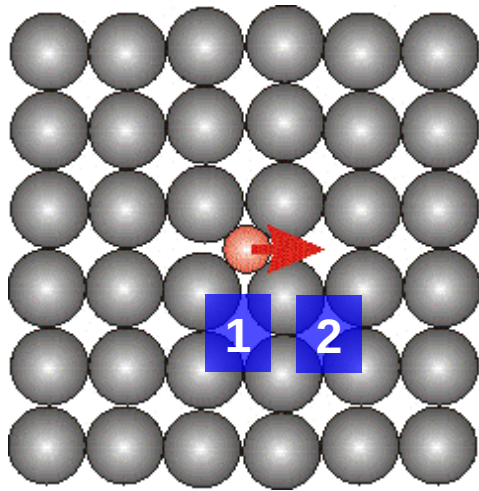
$$c(x,t) = \left(\frac{C_1 + C_2}{2} \right) - \left(\frac{C_2 - C_1}{2} \right) \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right)$$

Temperature dependence of diffusivity

$$D = D_0 e^{\left(-\frac{Q}{kT}\right)}$$

Atomic Models of Diffusion

1) Interstitial Diffusion



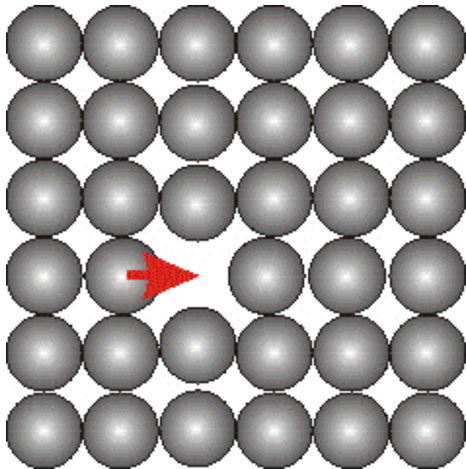
$$v' = v e^{\left(-\frac{\Delta H_m}{kT} \right)}$$

2) Substitutional diffusion via Vacancy Mechanism

$$\nu' = \nu e^{\left(\frac{-\Delta H_f}{kT}\right)} e^{\left(\frac{-\Delta H_m}{kT}\right)}$$

$$\nu' = \nu e^{\left(\frac{-\Delta H_f - \Delta H_m}{kT}\right)}$$

- $\Delta H_m \rightarrow$ enthalpy of motion of atom across the barrier.
- $\nu' \rightarrow$ frequency of successful jumps.



$$D = \nu' \delta^2$$

For Interstitial Diffusion

$$D = \nu \delta^2 e^{\left(\frac{-\Delta H_m}{kT}\right)} \quad \text{which is of the form}$$

$$D = D_0 e^{\left(\frac{-\Delta H_m}{kT}\right)}$$

- $D \text{ (C in FCC Fe at } 1000^\circ\text{C)} = 3 \times 10^{-11} \text{ m}^2/\text{s}$

For Substitutional Diffusion

$$D = \nu \delta^2 e^{\left(\frac{-\Delta H_f - \Delta H_m}{kT}\right)} \quad \text{which is of the form}$$

$$D = D_0 e^{\left(\frac{-\Delta H_f - \Delta H_m}{kT}\right)}$$

- $D \text{ (Ni in FCC Fe at } 1000^\circ\text{C)} = 2 \times 10^{-16} \text{ m}^2/\text{s}$

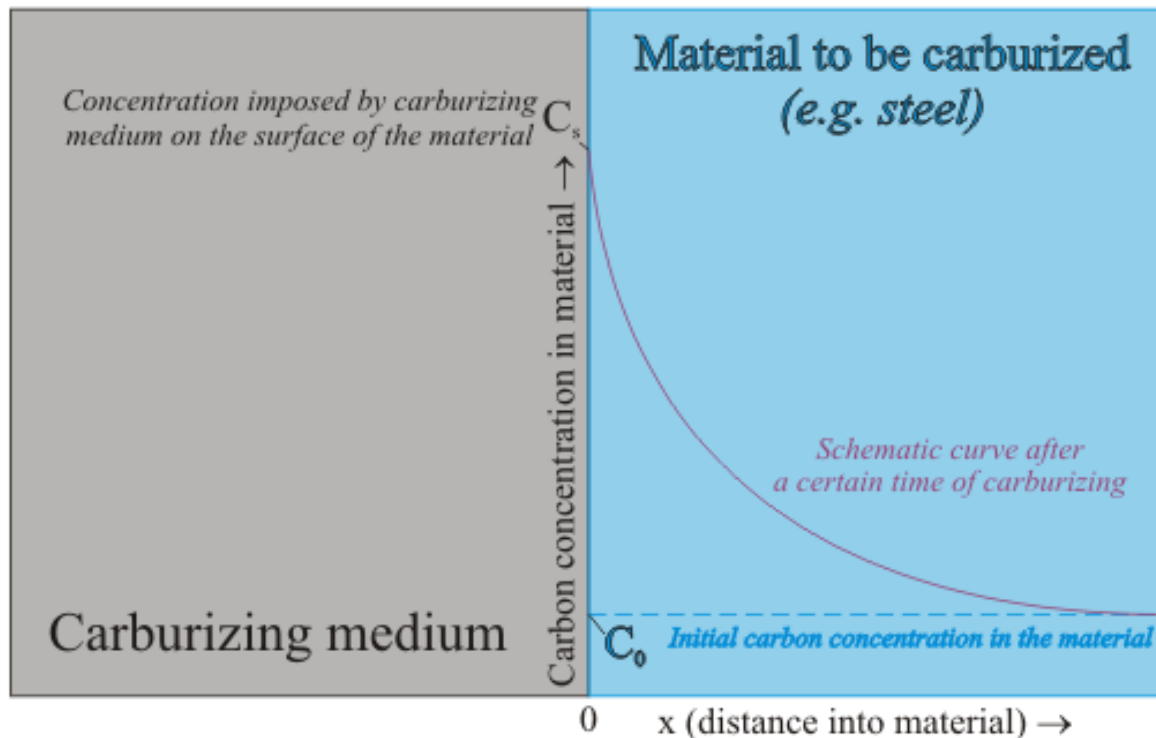
Diffusion Paths with Lesser Resistance

Experimentally determined activation energies for diffusion

$$Q_{\text{surface}} < Q_{\text{grain boundary}} < Q_{\text{pipe}} < Q_{\text{lattice}}$$

Applications based on Fick's II law → *Carburization of steel*

- ✓ In **pack carburizing**, a solid carbon powder used as C source.
- ✓ In **gas carburizing** Methane gas is used a carbon source using the following reaction.
 $\text{CH}_4 (\text{g}) \rightarrow 2\text{H}_2 (\text{g}) + \text{C}$ (the carbon released diffuses into steel).



We already have the error function solution to the diffusion equation.

$$C(x,t) = A - B \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Using the B.C. we can get the specific solution for the current case (i.e. the values of A & B).

- $C(+x, 0) = C_0$
- $C(0, t) = C_s$

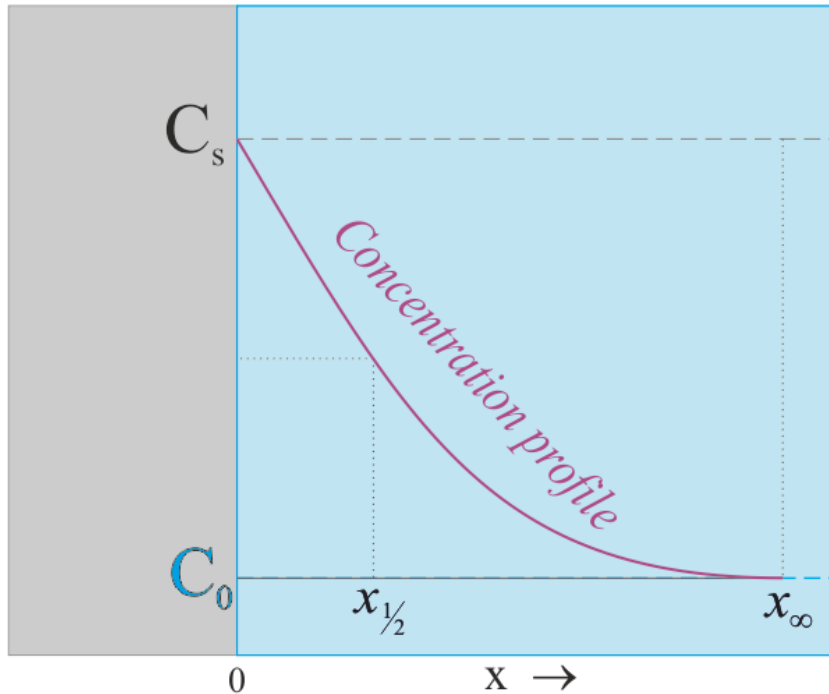


- $A = C_s$
- $B = C_s - C_0$

Hence, the solution is as below.

$$\frac{C(x,t) - C_s}{(C_0 - C_s)} = \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Approximate formula for depth of penetration



$$\left(\frac{C(x,t) - C_s}{C_0 - C_s} \right) = \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right) \quad \left(\frac{c(x,t) - C_0}{C_s - C_0} \right) = 1 - \operatorname{erf} \left(\frac{x}{2\sqrt{Dt}} \right)$$

Let the distance at which
 $[(C(x,t) - C_0)/(C_s - C_0)] = 1/2$ be called $x_{1/2}$
 (which is an 'effective penetration depth')

$$\frac{1}{2} = 1 - \operatorname{erf} \left(\frac{x_{1/2}}{2\sqrt{Dt}} \right) \quad \operatorname{erf} \left(\frac{x_{1/2}}{2\sqrt{Dt}} \right) = \frac{1}{2}$$

$$\operatorname{erf} \left(\frac{1}{2} \right) \approx \frac{1}{2} \Rightarrow \left(\frac{x_{1/2}}{2\sqrt{Dt}} \right) = \frac{1}{2}$$

$$x_{\text{penetration}} = \sqrt{Dt}$$

$$x_{1/2} = \sqrt{Dt}$$

The depth at which $C(x)$ is nearly C_0 is (i.e. the distance beyond which is 'un'-penetrated):

$$0 = 1 - \operatorname{erf} \left(\frac{x_\infty}{2\sqrt{Dt}} \right) \quad \operatorname{Erf}(u) \sim 1 \text{ when } u \sim 2 \quad \left(\frac{x_\infty}{2\sqrt{Dt}} \right) = 2 \quad x_\infty = 4\sqrt{Dt} \Rightarrow x_\infty = 4x_{1/2}$$

Diffusion in ionic materials

- ✓ Ionic materials are not close packed (like CCP or HCP metals).
- ✓ Ionic crystals may contain connected void pathways for rapid diffusion.
- ✓ These pathways could include ions in a sublattice (which could get disordered) and hence the transport is very selective
 - β alumina compounds show cationic conduction
 - Fluorite like oxides are anionic conductors.
- ✓ Due to high diffusivity of ions in these materials they are called superionic conductors. They are characterized by:
 - High value of D along with small temperature dependence of D
 - Small values of D_0 .
- ✓ Order disorder transition in conducting sublattice has been cited as one of the mechanisms for this behaviour.

Calculated and experimental activation energies for vacancy Diffusion

Element	ΔH_f	ΔH_m	$\Delta H_f + \Delta H_m$	Q
Au	97	80	177	174
Ag	95	79	174	184